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AI Engineering Blueprint for On-Premises Retrieval-Augmented Generation Systems

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Motivation

Retrieval-Augmented Generation (RAG) [1] extends LLMs with up-to-date, enterprise-internal knowledge. Yet GDPR, the EU AI Act, and HIPAA require on-premises deployment for sensitive data. A recent MIT study [2] finds that **only 5% of custom enterprise AI tools reach production**, blocked by integration complexity.

- Existing blueprints and reference architectures focus on **cloud** deployments.
- Open-source RAG implementations **lack enterprise-grade components** like access control, guardrails or observability.
- No comprehensive, code-level reference architecture for **on-premises** RAG.

Contributions

1. An **end-to-end reference architecture** for on-premises enterprise RAG, formally described using Kruchten's **4+1 view model** [3].
2. A **deployable reference application** as an adaptable starting point, including all enterprise-grade components.
3. **Best practices** for tooling, development, and CI/CD pipelines tailored to on-premises container deployment.

Code Paper

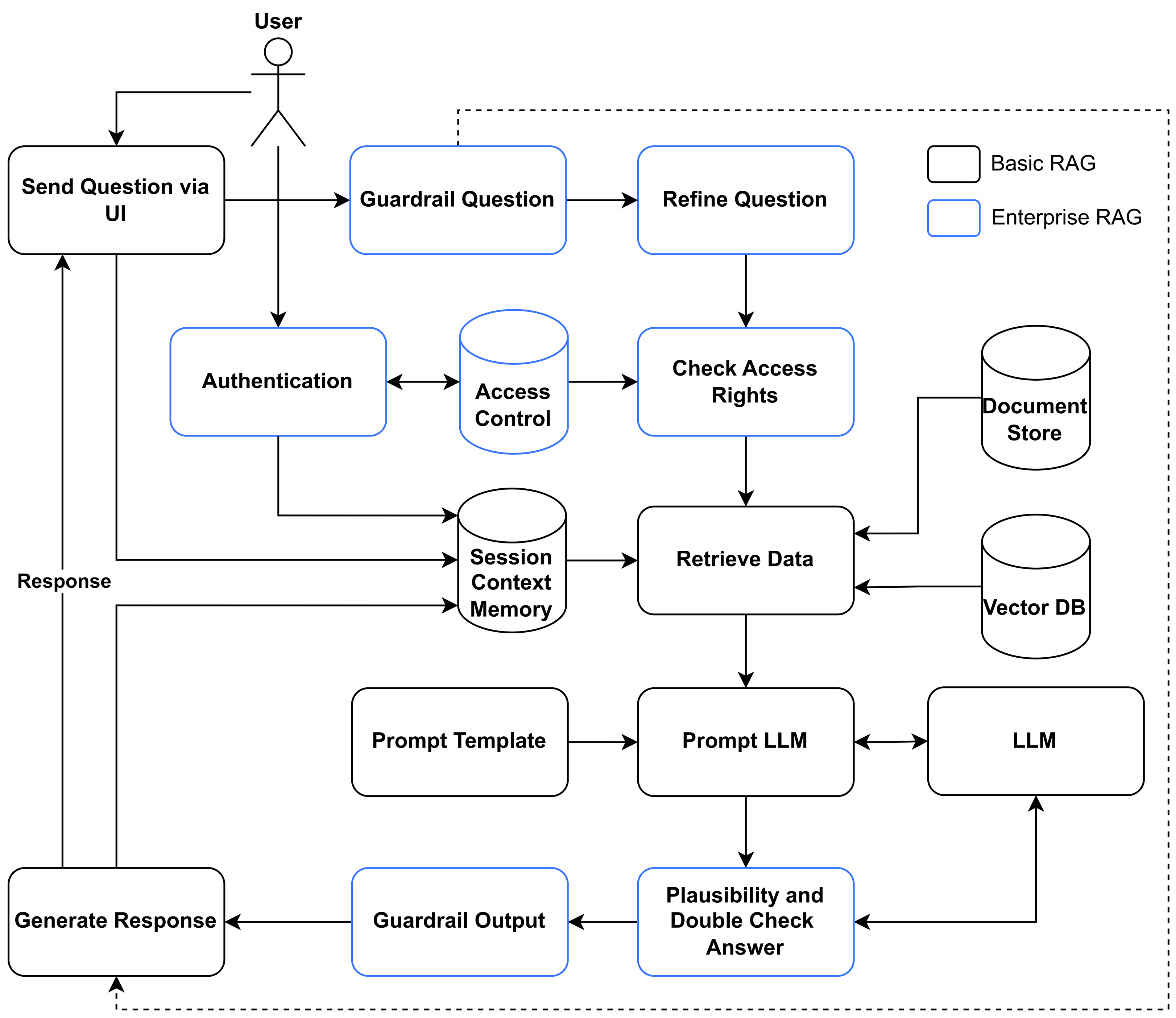
Requirements

The architectural design is informed by the requirements identified in the extant literature [4, 5, 6]. The following requirements are considered:

1. Security & Data Protection
2. Quality, Relevance & Accuracy
3. Explainability & Transparency
4. Performance
5. Continuous Learning
6. Continuous Operation
7. Integration in Setup
8. Scalability
9. Licensing & Copyright
10. Ethical Considerations & Bias

Functional Architecture

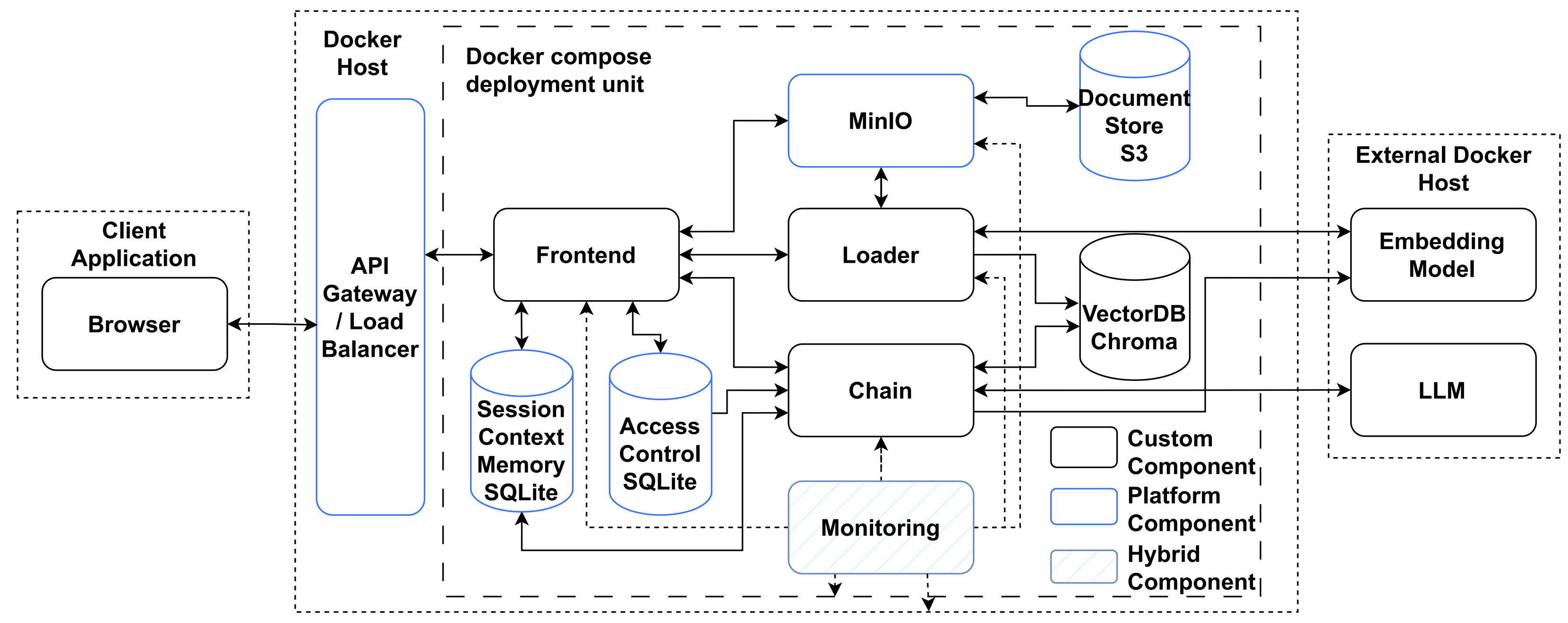
A **staged design** separates a basic RAG core (black) from **enterprise extensions** (blue), allowing organizations to incrementally adopt enterprise components based on their specific requirements.



Authentication and access control enforce per-user document visibility. Guardrails, query refinement and answer verification add quality and compliance guarantees at the cost of additional LLM calls.

On-Premises Deployment

The deployment view maps each component to an on-premises module. **Blue boxes** denote *platform placeholders* (e.g. MinIO, databases), designed to be substitutable with existing enterprise infrastructure without architectural changes.



Combined with locally hosted LLMs and embedding models, **all data processing remains within the enterprise IT infrastructure**.

Key Components

Frontend: test UI; exposes sources & chunks. Explainability & Transparency

Chain: retrieval and generation orchestration. Quality, Relevance & Accuracy

Loader: ingestion of enterprise data sources. Integration in Setup

Guardrails / Refinement / Verification: compliance gates. Quality, Relevance & Accuracy Ethical Considerations & Bias

Auth & Access Control: per-user document retrieval. Security & Data Protection

Monitoring: `OpenTelemetry` for logs, metrics, traces. Continuous Operation

Architectural Trade-offs

Decision	Benefit	Cost
Microservices and REST	independent scaling & replacement	minor network overhead
Extra LLM calls *guardrails, refinement, verification*	quality, compliance & reduced bias	response latency
Platform components	easy enterprise substitution	not performance-tuned
Docker Compose	easily adaptable trial deployment	limited resilience and scalability
Test Frontend	adaptable to existing enterprise UIs	not production-polished

Summary & Future Work

The blueprint operationalizes the **RAGOps** paradigm [6] by combining a formal reference architecture, a deployable reference application, and on-premises CI/CD pipelines.

Next steps:

- Industrial evaluation with expert interviews & case studies following DSR.
- Advanced RAG: Agentic RAG, multimodality, reranking.
- Evaluation metrics for retrieval, generation & system performance.
- Components for continuous learning and stronger explainability.

Goal: Lower the barrier for enterprises to adopt *on-premises* RAG in production.

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